

Industry and Topic: What Drives NBA Team Popularity?

Our project is centered around the business question: *How does fan sentiment relate to NBA team performance during the 2019–2020 season, and how accurately can team performance be predicted using historical team statistics?* To answer this question, this analysis investigates the relationship between on-court success and online fan engagement across NBA franchises. While successful teams are often expected to generate greater fan interest, fan sentiment may not always align perfectly with team performance. By combining NBA team performance statistics, social media sentiment data, and geographic information, this report examines whether teams with stronger performance also generate more positive fan sentiment and higher levels of engagement. In addition, a regression model trained on historical NBA team statistics from the 2009–2010 through 2018–2019 seasons is used to predict team wins during the 2019–2020 season. Through predictive modeling and data visualization, this analysis aims to identify patterns between fan sentiment, team performance, and prediction accuracy, providing insight into the factors that influence team popularity and success. The findings can help sports organizations better understand fan behavior, evaluate the relationship between sentiment and performance, and assess the effectiveness of data-driven approaches for forecasting team success.

Summary of Our Data/Datasets

The NBA statistics dataset contains team-level performance data, with each observation representing a team's statistics for a specific season. The key performance variables include wins, losses, win percentage, points scored, assists, rebounds, shooting percentages, three-point attempts, and plus/minus ratings. Since the datasets include NBA teams from cities across the United States and Canada, such as Los Angeles, Boston, Miami, Chicago, Dallas, and Toronto, the project also provides strong geographic coverage for regional analysis and visualization.

The NBA Teams dataset provides geographic information for all 30 NBA franchises and serves as the foundation for analyzing the league's geographic distribution across North America. The dataset includes each team's city, state or province, arena location, regional classification, and custom latitude and longitude coordinates, enabling the creation of geographic visualizations and maps. These attributes enable the examination of regional patterns, franchise distribution, market concentration, and the NBA's geographic footprint. The dataset is particularly valuable for map-based analyses because it links each franchise to a precise location, enabling visualization of team locations, regional groupings, arena characteristics, and the league's overall geographic structure.

The Twitter sentiment dataset consists of two NBA tweet datasets collected during the 2020 NBA season, covering tweets from approximately July 2020 through October 2020. The combined raw tweet data contains 116,825 tweet records, and after removing duplicate tweet IDs, 62,388 unique tweets

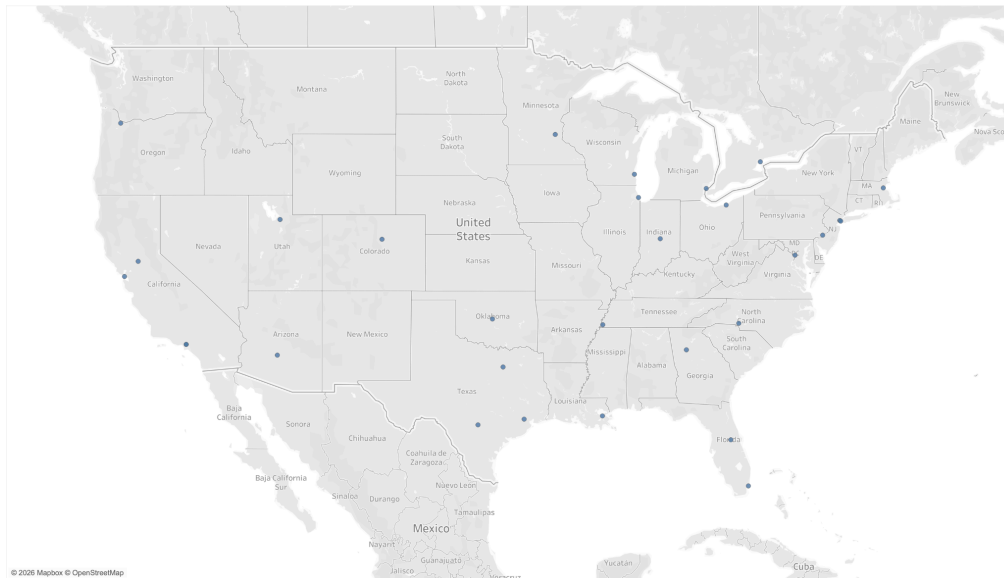
remained for analysis. Each observation represents one tweet related to an NBA team. Important variables include created_at, file_name, followers, friends, location, retweet_count, screenname, search_query, text, twitter_id, username, and polarity. The polarity variable is the primary sentiment score used in the visualizations: higher values indicate more positive sentiment, values near zero indicate neutral sentiment, and lower values indicate negative sentiment. For the visualizations, the tweet data was grouped by team to calculate each team's average and median sentiment scores, tweet count, positive, neutral, and negative tweet shares, total retweets, and average author follower count. These measures were used to compare online fan sentiment and engagement across teams and to link sentiment to team performance metrics, such as wins and win percentage, from the NBA team statistics dataset. One important limitation is that the Twitter sentiment dataset covers only a short 2020 period and does not treat all NBA teams equally. Therefore, this dataset is best used as a snapshot of online fan reaction during the 2020 season rather than a complete long-term measure of NBA fan sentiment.

The NBA Prediction Results dataset contains the outputs of a regression model developed to predict team win totals for the 2019–2020 NBA season. The model was trained using historical NBA team statistics from the 2009–2010 through 2018–2019 seasons, incorporating variables such as points scored, rebounds, assists, and other key performance metrics. The primary objective of the model was to evaluate how accurately team performance statistics could forecast regular-season wins in the 2019–2020 season. The dataset includes both actual and predicted win totals, enabling an assessment of model performance and prediction accuracy.

Overall, the key variables that will likely play an important role in our visualizations include win percentage, points scored, assists, shooting percentages, sentiment polarity scores, engagement metrics, follower counts, and team market locations. Initial analysis suggests that teams with stronger winning records and larger fan bases may generate more positive online sentiment, while underperforming teams may receive more negative online fan reactions. Additionally, historically popular teams may still maintain strong engagement and social media attention even during weaker seasons, suggesting that popularity and team performance may not always align directly.

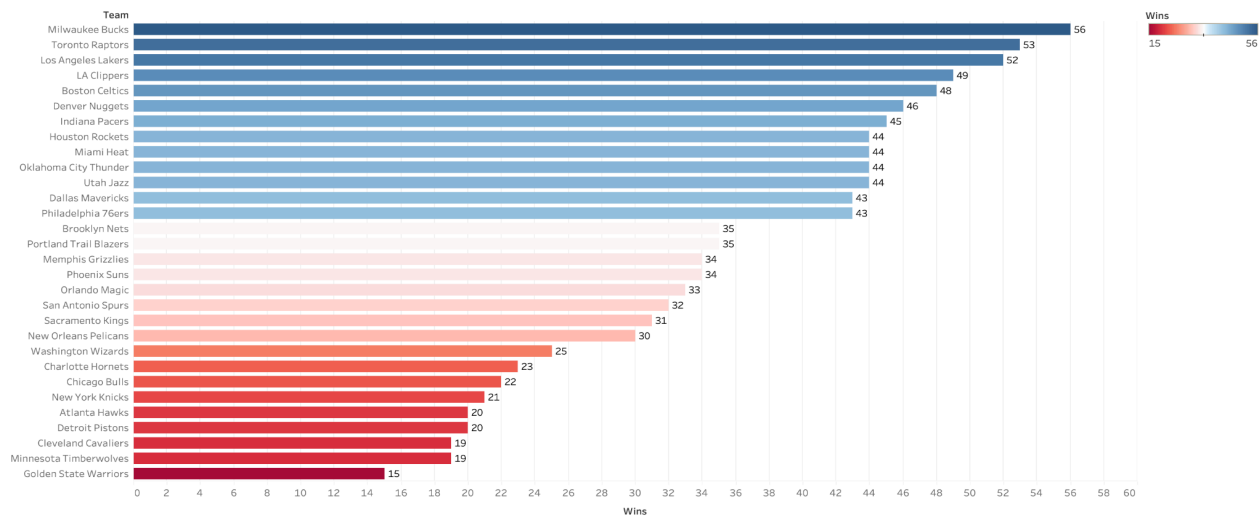
Storytelling Approach and Visualizations:

Team Locations Across the Country



Our first visualization is a map chart that focuses on visualizing the locations of each team in the NBA across the North American continent. Almost all of the teams in the NBA are located within the United States, but the one team that lies in Canada are the Toronto Raptors. This visualization maps each team across a map chart which gives the audience a chance to see where NBA teams exist geographically. Within Tableau, hovering over each point in the map chart provides information about the team and the arena where they play home games along with the geocoordinates of the team's city. The geographic distribution of NBA teams provides valuable context for understanding regional differences in fan engagement, media exposure, and franchise presence across North America. Additionally, this map illustrates the regional distribution of NBA franchises, showing where teams are located throughout the league.

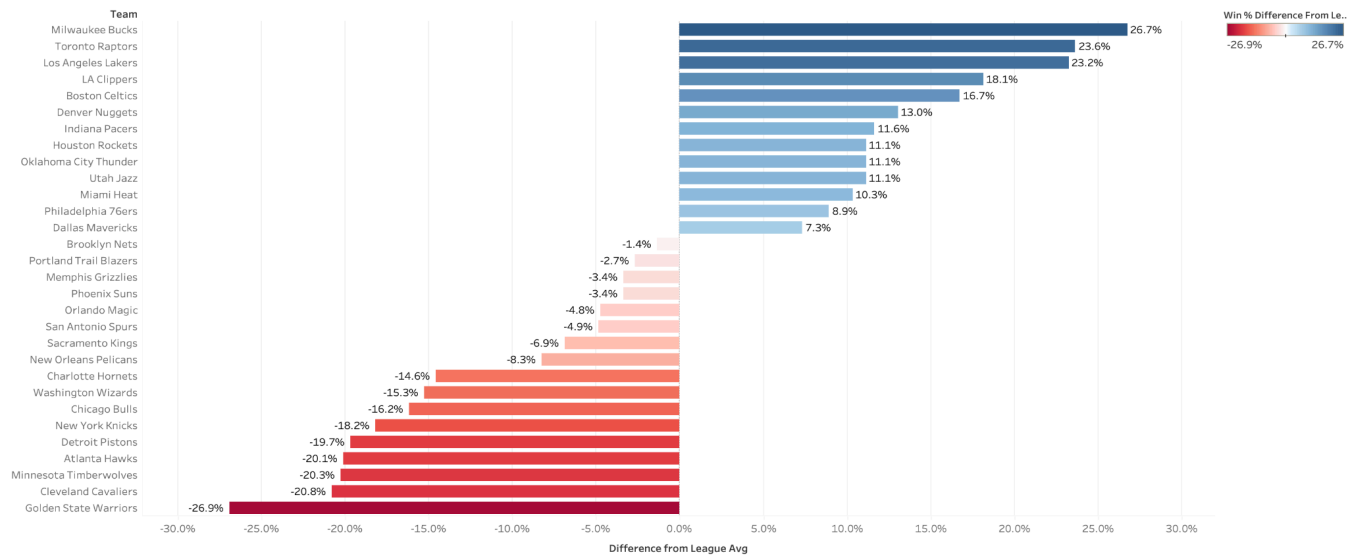
NBA Team Winning During the 2020 Season



First, let's gain an understanding of how each team in the NBA performed in the 2019/2020 season. Here, this visualization provides a league wide overview of team performance by comparing total regular season wins across all franchises. As the starting point of the analysis, it identifies which teams were most successful on the court before examining whether that success translated into stronger fan sentiment and engagement. The Milwaukee Bucks led the league with 56 wins, followed closely by the Toronto Raptors and Los Angeles Lakers, highlighting their dominance throughout the season. Other competitors, including the Los Angeles Clippers, Boston Celtics, and Denver Nuggets, also recorded strong performances with win totals above 45 games. At the opposite end of standings, teams such as the Golden State Warriors, Minnesota Timberwolves, and Cleveland Cavaliers struggled to generate the same level of success. The wide range of win totals demonstrates the significant differences in competitive performance across the league.

Team success is often viewed as a major driver of fan interest and overall engagement. Winning teams typically receive greater media attention, attract larger audiences, and generate more excitement among their fan bases. Strong performance can increase social media activity as fans discuss games, players, and championship expectations. For this reason, win totals provide an important benchmark for evaluating whether fan sentiment aligns with on-court success. As a result, some teams may remain highly popular despite lower win totals, while others may receive less attention despite strong records. This visualization establishes the competitive landscape of the NBA and serves as the foundation for examining the relationship between team performance and fan sentiment.

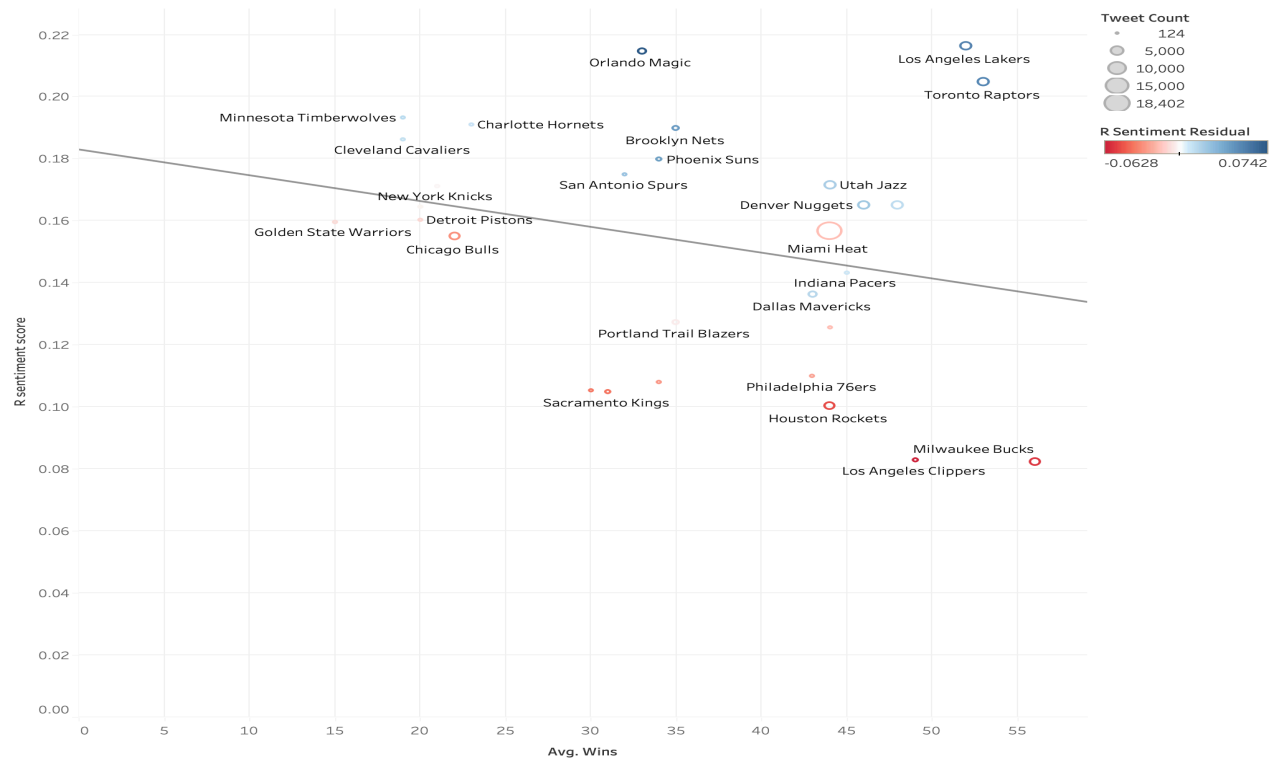
Team Win Percentage vs League Average



While the previous visualization compared total wins across teams, this chart adds important context by measuring each team's win percentage relative to the league average. Comparing teams to the league average helps account for differences in overall league competitiveness, since some seasons have stronger or weaker performance across the NBA as a whole. Rather than focusing only on raw win totals, this approach shows how far each team exceeded or fell below the typical level of success during the 2019-2020 season. The Milwaukee Bucks stood out as the strongest team, finishing 26.7% above the league average, while the Toronto Raptors and the Los Angeles Lakers significantly outperformed the rest of the league. On the other end of the spectrum, teams such as the Golden State Warriors, Cleveland Cavaliers, and Minnesota Timberwolves finished well below the average. This comparison highlights the gap between elite teams and struggling franchises more clearly than raw wins alone. By establishing which teams truly separated themselves from the league norm, this visualization provides a stronger foundation for later analyses exploring whether exceptional performance is associated with higher fan sentiment and engagement.

The results reveal that not all playoff teams performed at the same level, with a few teams clearly standing out from the rest. Teams near the league average highlight how competitive the middle of the standings was during the season. Looking at the performance relative to the league average provides more context than win totals alone. These differences are important because fan reactions may depend not only on winning, but also on how dominant a team is compared to its competitors.

Wins Vs Calculated Sentiment



Having established how teams performed relative to the rest of the league, this scatterplot investigates the relationship between team success and fan sentiment, by comparing each NBA team's 2019-2020 win total with its average Twitter sentiment score. The x-axis represents the number of wins recorded during the season, while the y-axis displays each team's average sentiment score derived from Twitter data collected during 2020. Higher sentiment scores indicate more positive fan reactions, whereas lower scores reflect less positive or more negative sentiment. Each point on the chart represents a single NBA franchise, allowing for direct comparison between on-court success and online fan perception.

The size of each circle point represents Tweet Count, which is the number of unique tweets collected for that team after duplicate tweet IDs were removed. Larger circles indicate higher levels of online engagement and discussion among the fans. Point color represents the sentiment residual, which measures how far a team's actual sentiment is from the sentiment predicted by the R model. Positive residuals signify the team received more positive sentiment than expected, while a negative residual means the team received less positive sentiment than expected. Together, these visual elements provide additional insight into both fan engagement and sentiment beyond simple win totals.

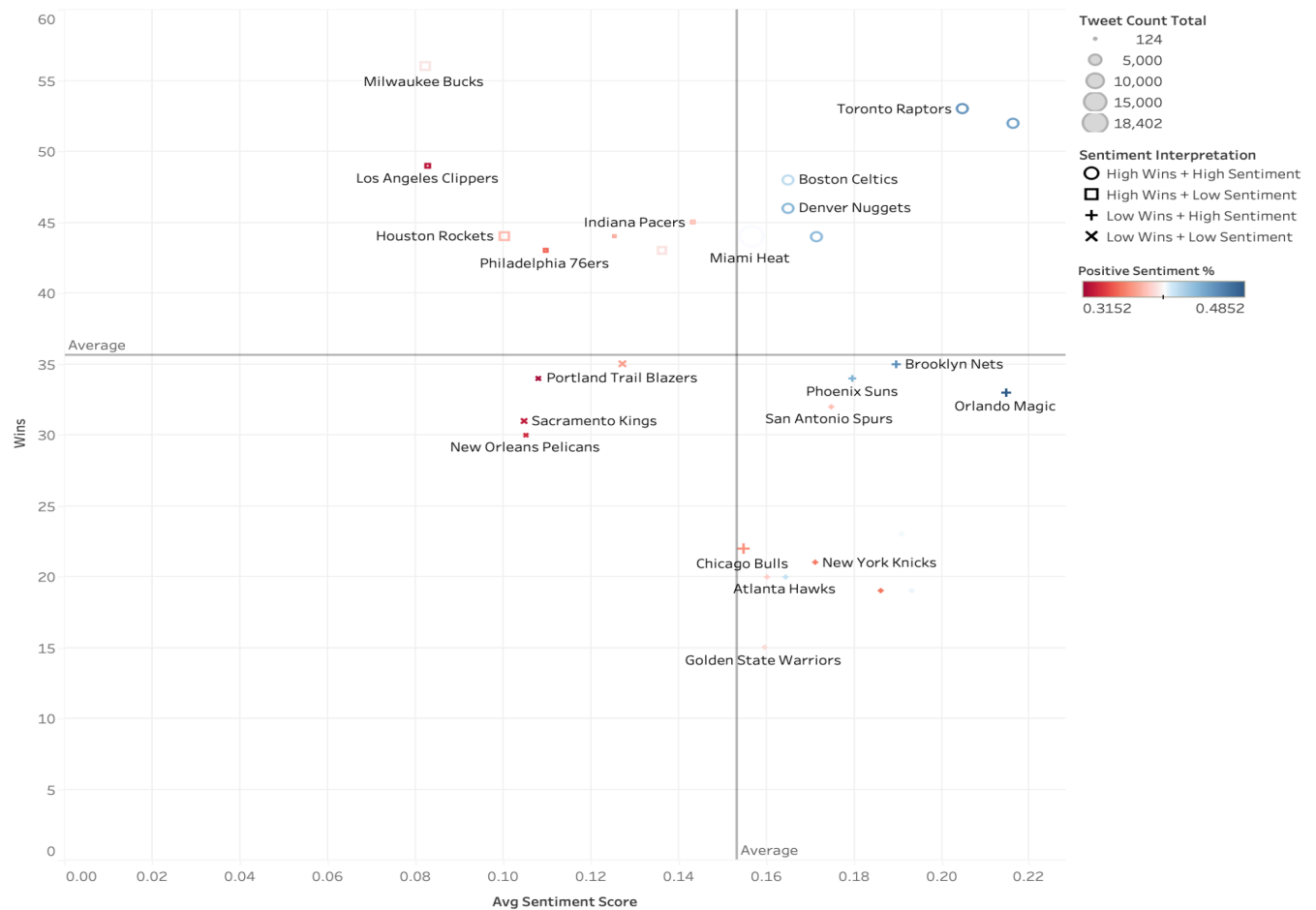
One of the primary goals of this analysis is to answer our main question: **Does winning drive NBA fan sentiment and online engagement?** The chart suggests that winning alone does not fully explain fan excitement. For instance, the Los Angeles Lakers and Toronto Raptors had high win totals and high sentiment scores, showing that strong performance can align with positive fan reaction. However, the

Milwaukee Bucks and Los Angeles Clippers also had high win totals but much lower sentiment scores, making them important outliers. Conversely, the Orlando Magic had fewer wins but one of the highest sentiment scores, showing that teams can receive positive online sentiment even without being one of the top-performing teams.

Differences in fan engagement are also evident throughout the chart. The Miami Heat generated the largest tweet volume, as shown by the largest circle, yet their sentiment score remained near the middle of the distribution. We can infer that high online engagement may reflect a larger or more active fan base rather than stronger approval or satisfaction. Fan conversations can be driven by a variety of components like expectations, roster changes, media attention and team narratives.

Several limitations should be considered when interpreting these results. The Twitter sentiment dataset captures only a portion of fan activity during a limited period in 2020 and should therefore be viewed as a snapshot rather than a comprehensive measure of long-term fan sentiment. In addition, tweet volume and coverage vary considerably across teams, meaning some franchises are represented more heavily than others. Despite these limitations, the visualization provides valuable insight into the complex relationship between team performance, fan sentiment, and online engagement. We see that this is highlighting that winning is only one of several aspects that shape how fans perceive and discuss NBA teams. Not all NBA teams are equally represented in the sentiment data and tweet volume varies widely throughout the teams.

Sentiment vs Performance

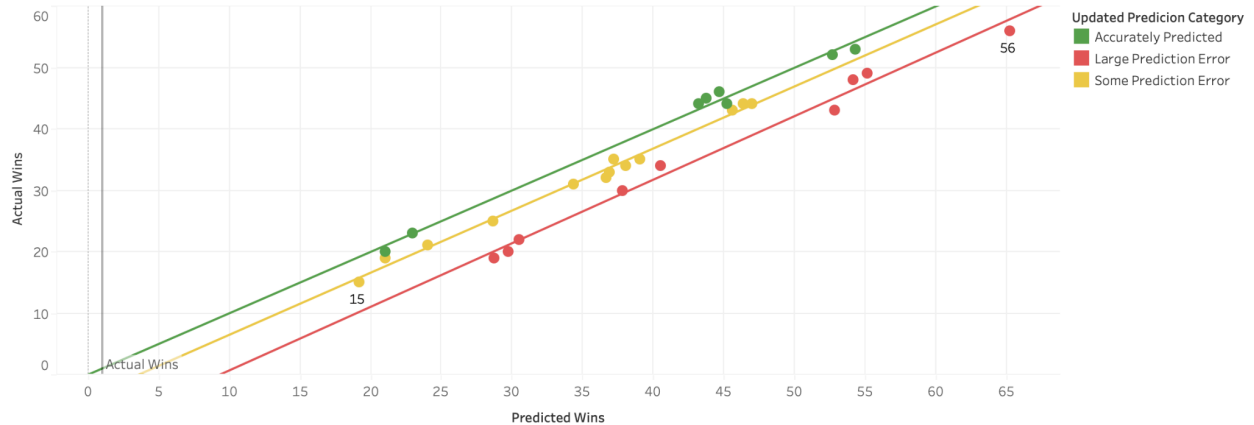


Building on the previous visualization, which compared team performance relative to the league average, this scatterplot explores whether stronger on-court performance is associated with more positive fan sentiment. Each point represents an NBA team, with wins displayed on the y-axis and average Twitter sentiment shows on the x-axis. The vertical and horizontal reference lines divide teams into four groups based on whether they are above or below average in wins and sentiment. Several teams, including the Toronto Raptors, Boston Celtics, and Denver Nuggets, appear in the upper-right quadrant, indicating both strong performance and positive fan sentiment. However, the chart also reveals that winning does not always guarantee the highest fan sentiment.

For example, the Milwaukee Bucks and Los Angeles Clippers recorded some of the highest win totals but had lower sentiment scores than several other successful teams. At the same time, teams such as the Orlando Magic and Brooklyn Nets generated relatively positive sentiment despite having below-average win totals. The size of each point represents tweet volume, showing that some teams generated substantial online discussion regardless of their performance level. Taken together, the results

suggest that while winning can influence fan sentiment, other factors such as franchise reputation, star players, team culture, and fan engagement also contribute to how teams are perceived.

2019-20 NBA Teams: Actual vs. Predicted Wins



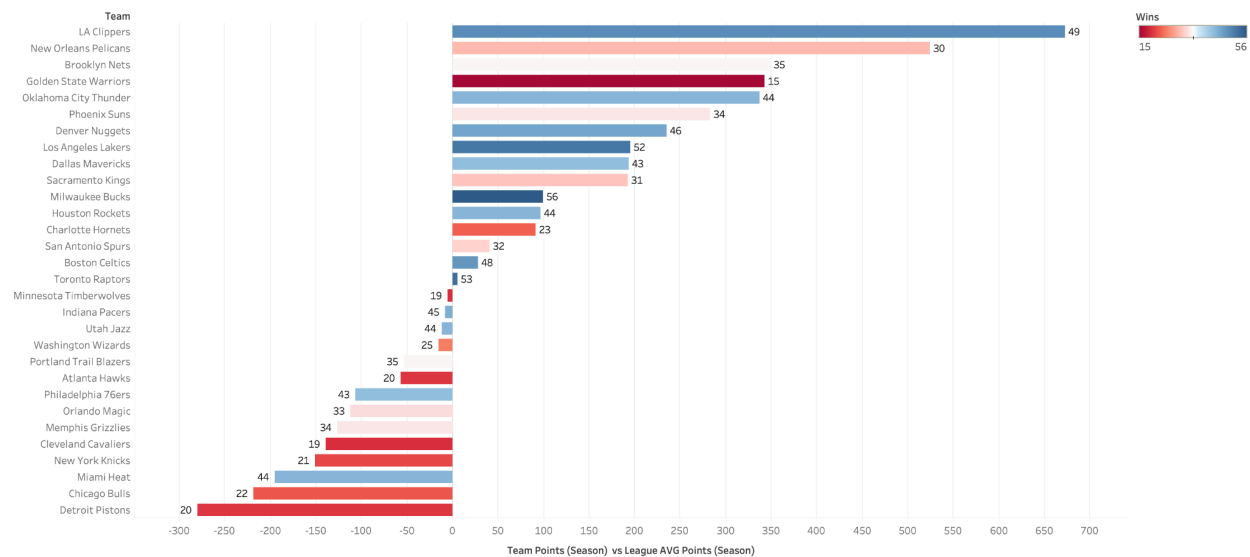
Prediction accuracy across all 30 NBA teams during the 2019-2020 season is illustrated in this scatterplot, with each point color-coded to the magnitude of prediction error: green represents accurately predicted teams within approximately 5 wins, yellow indicates moderate prediction error and represents larger prediction errors. The diagonal reference line represents perfect prediction, where actual wins equal predicted wins exactly. Overall, the regression model performed reasonably well, with most teams clustering near the line of perfect prediction and demonstrating a strong positive relationship between predicted and actual wins. Teams in the mid-range win totals between 30 and 45 wins, showed the most variability, with several falling into the yellow and red categories. This suggests that the model struggled most with predicting average-performing teams that may have overperformed or underperformed relative to historical trends. The two labeled outliers, representing teams with 15 and 56 actual wins, highlight the model's performance at the extremes. Here, we see that the team that finished with 56 wins was predicted to win 65 games, resulting in the largest prediction error in the dataset. These results suggest that while historical team statistics are a solid foundation for predicting wins, factors such as roster changes, injuries, and the unique circumstances of the 2020 bubble season likely contributed to prediction errors that a statistics-only model cannot fully capture.

Predicted Wins vs Actual Wins



To further evaluate the effectiveness of the prediction model, this scatterplot compares predicted wins against actual wins for all NBA teams during the 2019-2020 season. The x-axis shows the number of wins predicted by the regression model, while the y-axis represents the actual number of wins each team achieved during the season. Each point represents an NBA team and is grouped based on how accurate the prediction was. Green points indicate teams that were predicted accurately, yellow points represent teams with moderate prediction error and red points represent teams with larger predicted wins, generally corresponding to higher actual wins. This suggests that the model was effective at estimating team performance and that the green points appear across both strong and weaker teams, indicating that the model performed reasonably well for different levels of success. However, the red points show that some teams were harder to predict, likely because certain factors affecting performance were not included in the model. We can see that these results demonstrate that the regression model was able to provide useful win predictions while identifying areas where prediction accuracy could be improved.

Team Points Vs League Average (Wins as Color)



After examining team success from multiple perspectives, we evaluate scoring performance to see how it relates to team success. Here, this chart compares each team's total points scored against the league average. Teams displayed to the right of zero scored more points than the league average, while teams to the left scored fewer points than average. Color variation across the bars emphasizes differences in team win totals and overall performance. The Los Angeles Clippers stand out as the highest-scoring team relative to the league average, followed by the New Orleans Pelicans and Brooklyn Nets. Several successful teams, including the Los Angeles Lakers, Denver Nuggets, and Milwaukee Bucks, also scored well above the league average.

In contrast, teams such as the Detroit Pistons, Chicago Bulls, and the Miami Heat scored significantly fewer points than the league average. The chart demonstrates that many teams with higher scoring totals generally correlate with to record more wins, as implied by the darker blue bars. However, there are exceptions, suggesting that scoring alone does not fully determine team success. Factors such as defense, efficiency, and overall team balance may also contribute to winning games. Collectively, these results show that teams with stronger offensive production often achieved greater success during the season.

Report Summary and Concluding Remarks

Understanding what drives success in the NBA requires looking beyond wins and losses alone. This analysis explored the relationship between team performance, fan sentiment, and predictive modeling during the 2019-2020 season. The results convey that stronger teams often generated more positive fan sentiment and engagement, but winning alone did not fully explain fan reactions. Several teams received higher or lower sentiment scores than expected, suggesting that factors beyond on-court

performance influence how fans perceive their teams. The predictive model demonstrated that the team statistics can be used to estimate season performance with reasonable accuracy, although some prediction errors remained. Stronger offensive production was generally associated with greeted success during the season. Combining data visualization, sentiment analysis, and predictive analytics, it provided a deeper understanding of both team performance and fan behavior.

Technical Appendices (Included in Submission)

- R File containing code regarding dataset construction for our Prediction Charts
- Python File containing code regarding dataset construction for Sentiment Analysis Chart